

1. What is the greatest number of edges a graph with n vertices can have? Prove that any graph with this many edges has only one connected component.
2. Let G be a connected graph where all its vertices have degree 2, except for k of them, all of which have degree 1. What are the possible values of k ?
3. Let K_6 be the *complete graph on six vertices*, that is to say, the graph with six vertices with every pair of vertices connected by an edge. (In question 1, you determined how many edges this graph has, but you don't need that information for this problem.) Assume we color every edge either red or blue. Prove that, no matter how we do this, there exists a triangle (a set of three vertices, all connected by edges) with all the edges one color.
4. Find an example of a graph with five vertices that contains neither a clique with three vertices nor an independent set with three vertices. In other words, prove that Ramsey's theorem, as discussed in class, needs six vertices to work.
5. A *cycle* is a path that starts and ends at the same vertex, does not contain any other vertex more than once, and contains more than one vertex. Let G be a graph. Assume that there exist two vertices v, w of G , and assume that there are two different paths from v to w . Prove that G contains a cycle. *Hint:* It might be helpful to draw a diagram.